

Statistical Arbitrage via Pairs Trading: Algorithm Development, Mathematical Modelling, and Backtesting

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Introduction & Motivation

What is Pairs Trading?

- ▶ A **market-neutral statistical arbitrage strategy**.
- ▶ Exploits temporary divergence between two historically co-moving assets.
- ▶ A simultaneous long position in the underpriced asset and a short position in the overpriced asset.
- ▶ Profits once the spread reverts to its mean.

The Motivation

- ▶ Directional approaches (e.g., momentum trading) suffer severe drawdowns during market reversals.
- ▶ Pairs trading generates returns **independent of broad market movements**.
- ▶ Net exposure to the broader market is effectively cancelled out, preserving capital during volatility.

Project Objectives

1. **Machine Learning Pipeline:** Design an unsupervised ML pipeline for pair selection using PCA and OPTICS clustering.
2. **Mathematical Modelling:** Derive and implement four distinct trading strategies from first principles:
 - ▶ Bollinger Bands
 - ▶ Ornstein-Uhlenbeck (OU) stochastic process
 - ▶ Copula-based conditional probability
 - ▶ Cointegration
3. **Backtesting Engine:** Build a realistic out-of-sample backtesting engine accounting for hedge-ratio-based position sizing and transaction costs.
4. **Empirical Validation:** Compare all strategies across multiple pairs against a SPY buy-and-hold benchmark.

Data Preprocessing & Dimensionality Reduction

Data Collection & Cleaning

- ▶ **Universe:** S&P 500 constituents (Daily OHLCV from Stooq).
- ▶ **Filtering:** Tickers covering fewer than 80% of calendar trading days in the training window are excluded.
- ▶ **Standardisation:** Daily log-returns are computed and standardised (zero mean, unit variance) to place all assets on equal footing.

Dimensionality Reduction (PCA)

- ▶ The raw return matrix is high-dimensional and noisy.
- ▶ PCA projects it onto the directions of maximum variance.
- ▶ **Variance Threshold:** Retaining 80% of cumulative explained variance.
- ▶ **Outcome:** Retains dominant market factors while discarding idiosyncratic noise prior to clustering.

Density-Based Clustering (OPTICS)

The OPTICS Algorithm

- ▶ Orders data points by **reachability distance**.
- ▶ Particularly well-suited for financial data because it does not require the number of clusters to be pre-specified.
- ▶ Handles density-heterogeneous data effectively; valleys in the reachability plot correspond to dense clusters.

Pipeline Efficiency & Results

Using `min_samples = 3`, OPTICS identified **5 clusters** from the S&P 500 universe. This clustering yielded **34 candidate pairs** — representing a massive 97% reduction from the unconstrained pair space, making downstream statistical testing computationally viable.

Statistical Validation Filters

The Four Filters

Each candidate pair must pass all tests:

- **Cointegration (Engle-Granger):** ADF test on OLS spread residual ($p < 0.05$).
- **Hurst Exponent:** $\hat{H} < 0.5$ confirms anti-persistence (mean reversion).
- **Half-Life:** $1 \leq t_{1/2} \leq 252$ days ensures actionable trading speed.
- **Mean Crosses:** Minimum of 12 zero-crossings per year.

Year	Pair	Asset A	Asset B	Coint. p	$\hat{\beta}$	Hurst H	Half-life (d)	Mean X/yr
2022	AEP/CNP	American Electric Power	CenterPoint Energy	0.0023	0.294	0.322	6.8	35.9
	AEE/AEP	Ameren Corporation	American Electric Power	0.0082	1.029	0.265	6.2	33.1
	AWK/ATO	American Water Works	Atmos Energy	0.0229	0.457	0.403	8.0	21.2
2023	AWK/CNP	American Water Works	CenterPoint Energy	0.0124	0.080	0.393	7.2	21.4
	AEP/CNP	American Electric Power	CenterPoint Energy	0.0316	0.156	0.379	9.4	27.9
	LNT/AWK	Alliant Energy	American Water Works	0.0367	4.357	0.408	8.0	28.8
2024	APO/BAC	Apollo Global Management	Bank of America	0.0124	0.135	0.404	8.1	29.8
	ARES/BAC	Ares Management	Bank of America	0.0364	0.163	0.429	9.0	27.0

Trading Strategies I: Bollinger Bands & Cointegration

Bollinger Band Strategy

- ▶ Computes rolling mean and standard deviation of the spread (window $w = 20$).
- ▶ Defines a rolling z-score: $\bar{z}_t$.
- ▶ **Signal Rule:**
 - ▶ **Short:** If spread above upper band (> 2).
 - ▶ **Long:** If spread below lower band (< -2).
 - ▶ **Exit:** When spread crosses the mean.

Cointegration Strategy

- ▶ Uses a rolling OLS regression to track an evolving equilibrium (dynamic hedge ratio).
- ▶ **KPSS Guard:** The resulting spread is tested daily with the KPSS test. If non-stationary, positions are liquidated.
- ▶ **Signal Rule:** Z-score signal used for entries ($|z| > 2$) and exits ($|z| < 0.5$).

Trading Strategies II: Ornstein-Uhlenbeck (OU) Model

Continuous-Time Stochastic Model

The spread is explicitly modelled as a mean-reverting Ornstein-Uhlenbeck process:

$$dZ_t = \kappa(\theta - Z_t)dt + \sigma dW_t$$

where θ is the long-run mean, $\kappa > 0$ is the speed of mean reversion, and $\sigma > 0$ is the volatility.

Calibration & Execution

- ▶ The exact discrete transition is solved analytically.
- ▶ Parameters (κ, θ, σ) are estimated via Maximum Likelihood Estimation (L-BFGS-B minimisation).
- ▶ An equilibrium z-score s_t is calculated, grounding the entry and exit thresholds directly in the stationary distribution of the OU process.

Trading Strategies III: Copula Approach

Theoretical Foundation

- ▶ Spread-based frameworks assume linear dependence. Copulas capture the **full nonlinear dependence structure** between asset returns.
- ▶ Utilises **Sklar's Theorem** to factor the joint distribution, encoding dependence separately from the marginals.

Implementation & Signal Rule

- ▶ Implements three Archimedean families: **Clayton, Gumbel, and Frank**.
- ▶ The best-fitting family is selected daily by AIC on a rolling window.
- ▶ Trading signals are derived theoretically from conditional probabilities (p^A, p^B) rather than raw spread z-scores. Entry triggered when an asset is undervalued while the other is overvalued.

Backtesting Framework

Engine Specifications

- ▶ **Capital & Sizing:** \$100,000 starting capital. Both legs of the trade are sized explicitly using the estimated hedge ratio.
- ▶ **Friction:** Transaction costs of 5 bp per leg and slippage of 2 bp per trade are deducted.
- ▶ **Strict Out-of-Sample:** All parameters are estimated exclusively on the preceding 12-month training window. No look-ahead bias is introduced.

Multi-Regime Evaluation

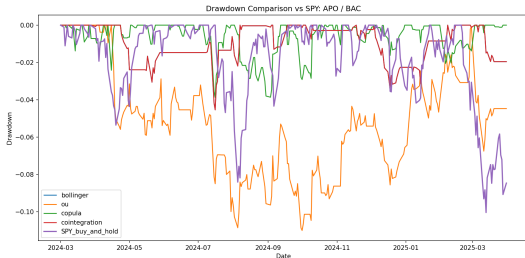
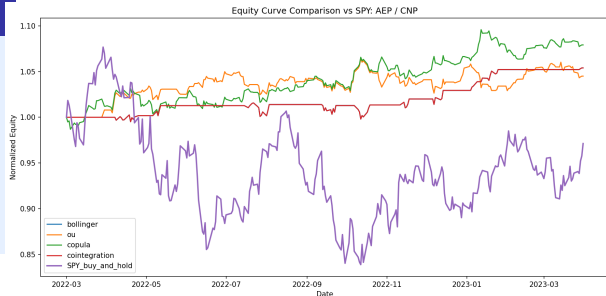
The strategies are evaluated across three deliberately contrasting market regimes to test strategy robustness:

1. **Test Window 1:** March 2022 – March 2023 (Bear Market)
2. **Test Window 2:** March 2023 – March 2024 (Bull Market)
3. **Test Window 3:** March 2024 – March 2025 (Strong Bull Market)

Results — Bear Market (Mar 2022 – Mar 2023)

Stress Test of Market Neutrality

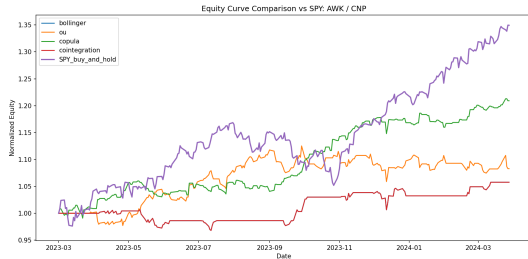
- **Benchmark:** SPY suffered a cumulative return of -2.87% with a massive max drawdown of -22.09%.
- **Performance:** All strategies across pairs (e.g., AEP/CNP) generated positive returns ranging from +4% to +19.5%.



Results — Bull Markets (2023 & 2024)

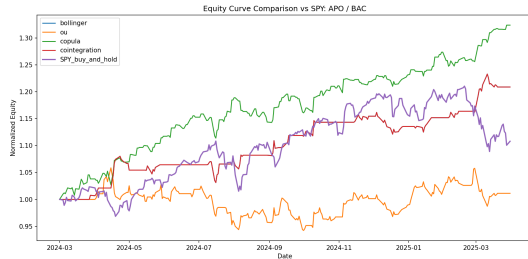
2023: Preserving Capital

- ▶ SPY rallied heavily (+34.94%).
- ▶ Copula still delivered $\approx 21\%$ absolute return. Bollinger and Cointegration earned 0-6%, confirming successful capital preservation without directional exposure.



2024: Copula Dominance

- ▶ SPY returned +10.81%.
- ▶ The Copula strategy on APO/BAC delivered an impressive **+32.39%**, outperforming SPY by roughly 10 points with near-zero drawdown.



Cross-Year Summary & Discussion

Key Takeaways

- ▶ **Copula Dominance:** The strategy is robust across regimes. It generates the highest absolute returns in all three years—including outperforming SPY outright in 2024—with consistently low drawdowns. Its adaptive rank-based signal is the key differentiator.
- ▶ **Capital Preservation:** Bollinger and Cointegration are capital-preserving, not alpha-generating, in bull markets. They deliver near-zero returns when SPY is strongly positive, successfully achieving the primary goal of a market-neutral strategy for an investor with other long-only positions.
- ▶ **No Absolute Losses:** No strategy lost money across any year. Even the worst case (-1.49%) was smaller than SPY's concurrent drawdown. This validates the capital-preservation thesis of the market-neutral design.
- ▶ **Adaptive ML Pipeline:** The shift from utility pairs (2022–2023) to financials (2024) demonstrates that the PCA + OPTICS pipeline correctly identifies dominant co-moving clusters without manual intervention.

Performance Summary Across Market Regimes

Test Year	Strategy	Cum. Ret.	vs SPY	SPY Ret
2022 (bear)	Copula	+9% to +20%	beat	-2.87%
	OU	+4% to +8%	beat	
	Bollinger	-1% to +5%	beat or match	
	Cointegration	-1% to +5%	beat or match	
2023 (bull)	Copula	+20.92%	trail SPY (+34.94%)	+34.94%
	OU	+8.29%	trail	
	Bollinger	+5.77%	trail	
	Cointegration	+5.77%	trail	
2024 (bull)	Copula	+32.39%	beat SPY (+10.81%)	+10.81%
	Bollinger	+20.89%	beat	
	Cointegration	+20.89%	beat	
	OU	+1.13%	trail	

Conclusion

- ▶ **End-to-End Pipeline:** Developed a completely automated pipeline covering S&P 500 universe management, clustering, statistical validation, and backtesting.
- ▶ **Consistent Profitability:** Evaluated across three severely distinct market regimes—**no strategy produced a negative cumulative return in any year.**
- ▶ **Mathematical Edge:** The Copula's richer dependence modelling delivered superior performance compared to linear distance metrics in every test window evaluated.

Open-Source Implementation:

github.com/ImaginedTime/Pairs_trading_BTP2

Thank You!